

Autoencoders in a nutshell

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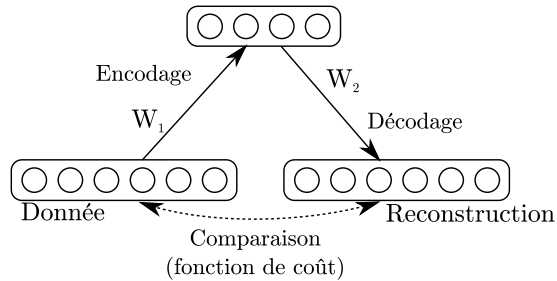
<https://velcin.github.io>

Formation UdL 2025-2026

Autoencoder

- What's the objective of AE?
 - extract low-dimensional representation
- How ?
 - reconstruct (decode) data from the internal (latent) code
- Some features
 - PCA can be seen as a linear AE
(see: <https://arxiv.org/pdf/1804.10253.pdf>)
 - possibility to stack encoding layers
 - improve the latent space (c.f., VAE)

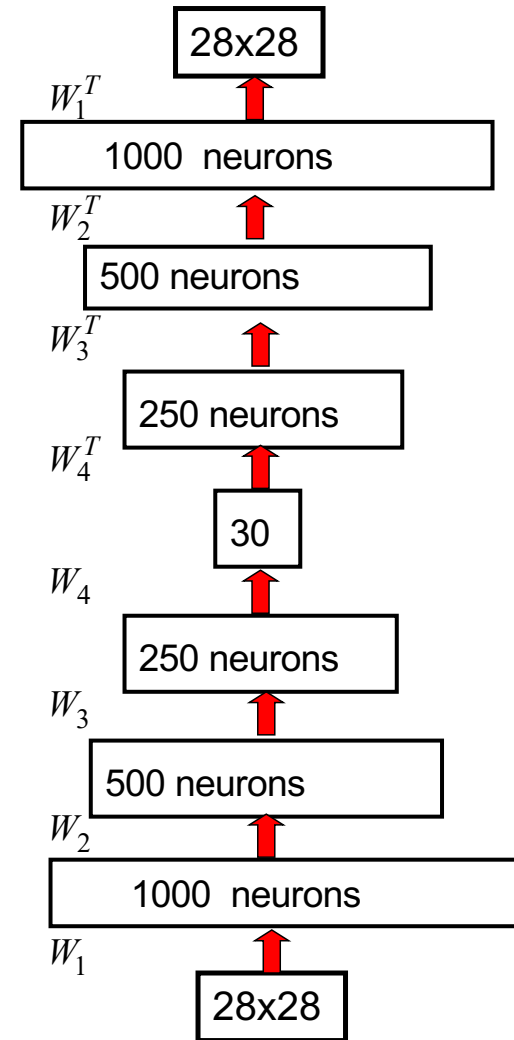
Architecture of AE (1)



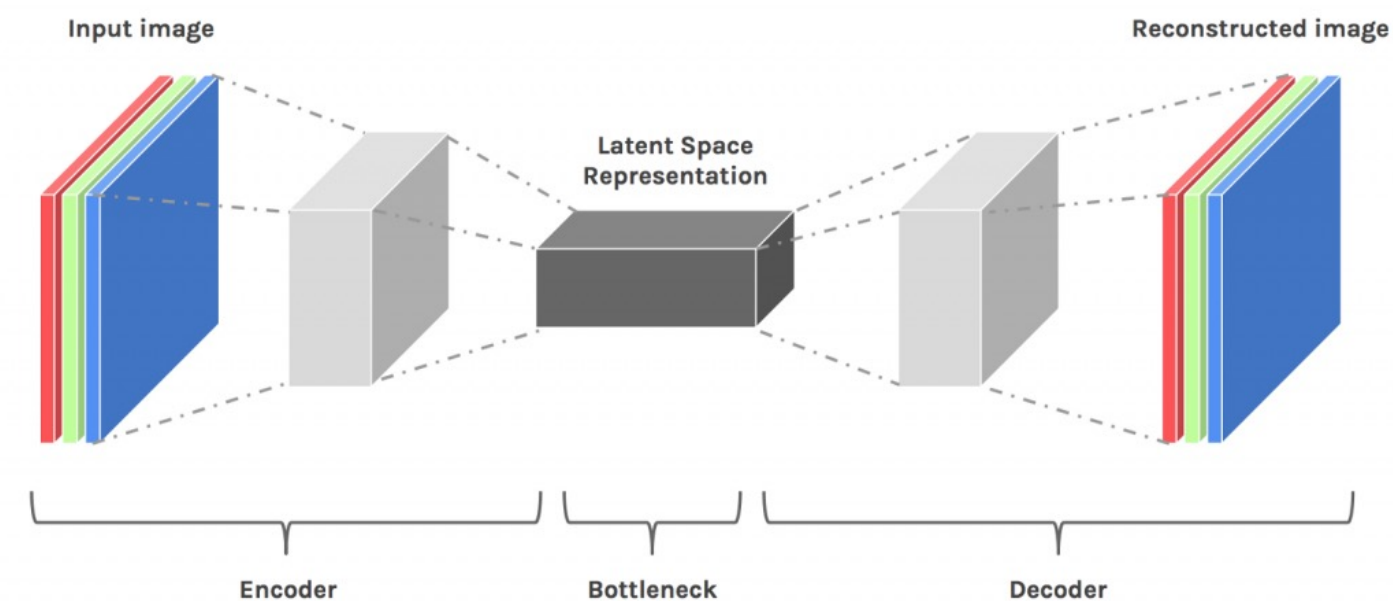
encoder : $p_{\text{encode}}(h|x)$

decoder: $p_{\text{decode}}(x|h)$

make it deep ->

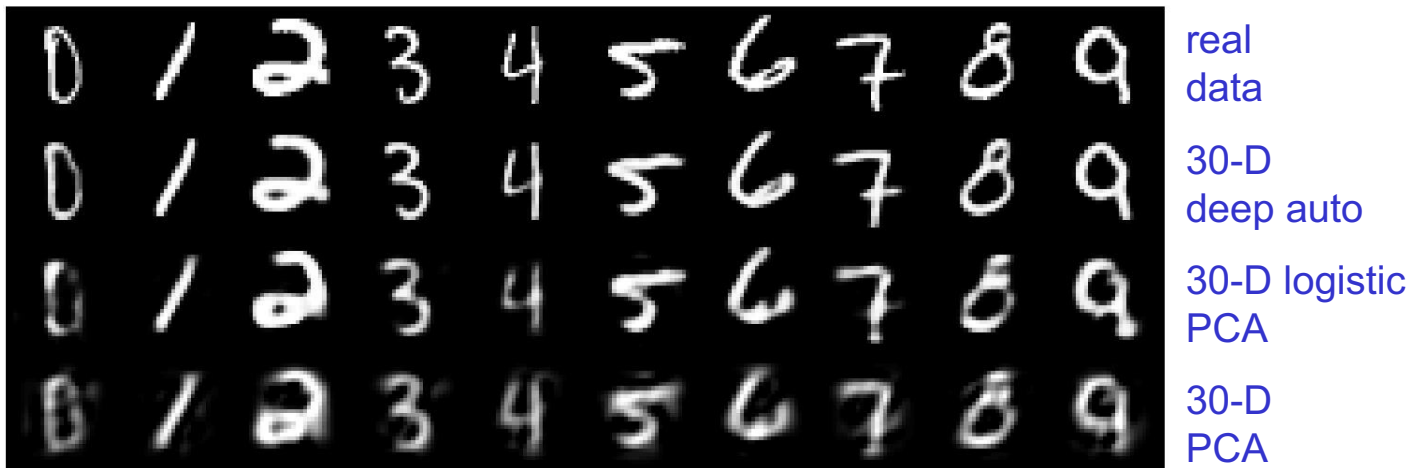


Architecture of AE (2)

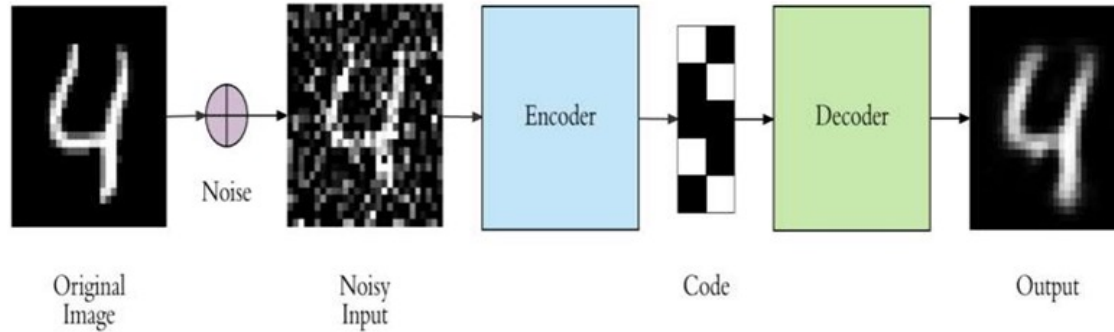


Comparison

taken from CSC2535: 2013 (G. Hinton)



Denoising AE



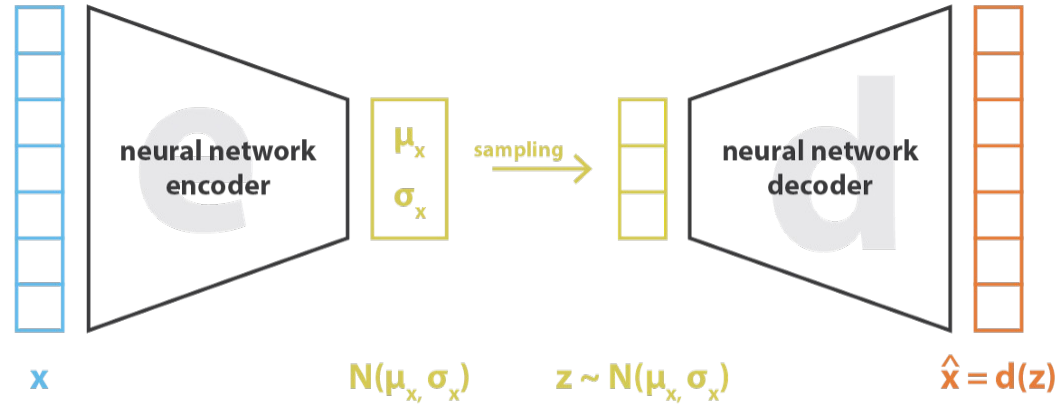
Sparse autoencoder

Adding a penalty on the code $\mathbf{h}$:

$$L(x, g(f(x))) + \Omega(h)$$

For instance, $\Omega(h) = \lambda \sum_i |h_i|$

Variational AE



$$\text{loss} = ||x - \hat{x}||^2 + \text{KL}[N(\mu_x, \sigma_x), N(0, I)] = ||x - d(z)||^2 + \text{KL}[N(\mu_x, \sigma_x), N(0, I)]$$

VAE (con't)

- Solution to get better latent spaces therefore better generation
- Two parts of the architecture:
 - encoder: $p(z|x)$
 - decoder: $p(x|z)$
- Optimization based on Variational Inference (VI)
- Implemented with the « reparametrization trick »
- The internal code (z) should follow a « simple » law (e.g., Gaussian)

VAE (con't)

